Tail Risk in SPY and XLF: GJR-GARCH vs EGARCH

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Abstract

This report evaluates asymmetric volatility models for tail-risk management using daily log returns of SPY and XLF from 2016-01-01 to 2026-01-01. Two models, GJR-GARCH(1,1) and EGARCH(1,1), are estimated on an in-sample period (80%) and assessed by rolling one-step-ahead out-of-sample forecasts (20%). Dynamic Value-at-Risk (VaR) and Expected Shortfall (ES) are computed at 95% and 99% confidence levels under Normal innovations, followed by exception-rate backtesting. The evidence shows strong volatility clustering, negative skewness, and heavy tails in both assets. At the extreme 99% confidence level, GJR-GARCH delivers tighter exception-rate alignment than EGARCH for both SPY and XLF, indicating more reliable tail coverage in severe market conditions.

1 Introduction & Executive Summary

Tail-risk management is central to financial institutions because solvency stress is driven by low-probability, high-impact losses rather than average market fluctuations. In practice, volatility is time-varying and clustered; moreover, downside shocks typically generate larger volatility responses than upside shocks of similar magnitude. This asymmetry, commonly referred to as the leverage effect, is critical for market-risk systems that must remain robust during crisis regimes.

The asset selection is intentionally macro-to-sectoral. SPY (S&P 500 ETF) proxies broad U.S. market risk and captures the aggregate macro-financial state. XLF (Financial Select Sector SPDR Fund) concentrates exposure to banks and diversified financial institutions, a segment with strong procyclicality, balance-sheet sensitivity, and systemic relevance. Comparing SPY and XLF therefore reveals whether asymmetric volatility models scale consistently from broad market risk to a shock-amplifying sector.

The key business conclusion is straightforward. Both models fit the data well, but under the stricter 99% tail standard used in supervisory and internal stress governance, GJR-GARCH produces better calibrated exception rates than EGARCH for both SPY and XLF in the out-of-sample period. This makes GJR-GARCH the preferred baseline for extreme-loss monitoring in this sample, while EGARCH remains competitive for in-sample fit metrics and 95% coverage in selected cases.

2 Theoretical Framework

2.1 Return and Error Decomposition

Let daily return be

$$r_t = \mu_t + \epsilon_t, \tag{1}$$

with innovation

$$\epsilon_t = \sigma_t z_t, \tag{2}$$

where σ_t is conditional volatility and z_t is an i.i.d. standardized shock. Under baseline assumptions, $z_t \sim \mathcal{N}(0, 1)$; a Student-$t$ alternative can be used to model heavier tails.

2.2 GJR-GARCH(1,1)

The GJR-GARCH variance recursion is

$$\sigma_t^2 = \omega + (\alpha + \gamma I_{t-1})\epsilon_{t-1}^2 + \beta\sigma_{t-1}^2, \tag{3}$$

where

$$I_{t-1} = \begin{cases} 1, & \text{if } r_{t-1} < 0, \\ \\ 0, & \text{if } r_{t-1} \geq 0. \end{cases} \tag{4}$$

Interpretation is immediate. For positive-return days, the shock-loading term is α; for negative-return days, it is $\alpha + \gamma$. A positive and significant γ quantifies a stronger volatility response to bad news and operationalizes downside asymmetry.

2.3 EGARCH(1,1)

The EGARCH specification is

$$\ln(\sigma_t^2) = \omega + \alpha \left(|z_{t-1}| - \mathbb{E}|z_{t-1}| \right) + \gamma z_{t-1} + \beta \ln(\sigma_{t-1}^2). \tag{5}$$

Because the model evolves in log-variance space, positivity of σ_t^2 is guaranteed without imposing non-negativity constraints on ARCH coefficients. The signed term γz_{t-1} captures directional asymmetry: when $\gamma < 0$, negative shocks increase volatility more than positive shocks.

2.4 Dynamic VaR and ES

For left-tail risk at confidence level $c \in \{0.95, 0.99\}$, define tail probability $\alpha = 1 - c$. Let q_α denote the α-quantile of z_t. Then one-step-ahead conditional VaR is

$$\mathrm{VaR}_{c,t+1} = \mu_{t+1} + \sigma_{t+1} q_\alpha. \tag{6}$$

For Normal innovations, $q_\alpha = \Phi^{-1}(\alpha)$.

Expected Shortfall is the conditional expectation beyond VaR:

$$\mathrm{ES}_{c,t+1} = \mathbb{E}\left[r_{t+1} \mid r_{t+1} \leq \mathrm{VaR}_{c,t+1} \right]. \tag{7}$$

Under Normality,

$$\mathrm{ES}_{c,t+1} = \mu_{t+1} - \sigma_{t+1} \frac{\phi\left(\Phi^{-1}(\alpha)\right)}{\alpha}, \tag{8}$$

where $\phi(\cdot)$ and $\Phi(\cdot)$ are the standard Normal pdf and cdf. In implementation, μ_{t+1} is set to zero for conservative and stable short-horizon risk forecasting.

3 Data & Exploratory Data Analysis

3.1 Sample and Construction

The dataset contains daily log returns (in percentage points) for SPY and XLF from 2016-01-01 to 2026-01-01, totaling 2,513 observations per series. Prices are downloaded via Yahoo Finance and transformed as

$$r_t = 100 \times \left(\ln P_t - \ln P_{t-1}\right). \tag{9}$$

The sample includes two major stress windows highlighted in the time-series plot: the COVID shock (March 2020) and the U.S. regional banking stress episode around Silicon Valley Bank (March 2023).

3.2 Descriptive Statistics

Table 1 confirms non-Gaussian return behavior: both assets are negatively skewed and strongly leptokurtic.

Table 1. Descriptive Statistics of Daily Log Returns (%)

Asset	Mean	Std	Skewness	Kurtosis	Min	Max	Obs
SPY	0.0552	1.1369	-0.6048	18.1541	-11.5887	9.9863	2513
XLF	0.0496	1.4087	-0.5880	18.5290	-14.7448	12.3602	2513

XLF exhibits higher volatility and deeper downside realizations than SPY, consistent with stronger cyclicality and higher crisis sensitivity in the financial sector.

4 Empirical Results & Model Estimation

4.1 In-Sample Estimation Design

The first 80% of observations (roughly 2016 through end-2023) are used for parameter estimation. Both GJR-GARCH(1,1) and EGARCH(1,1) are estimated with Normal innovations using maximum likelihood.

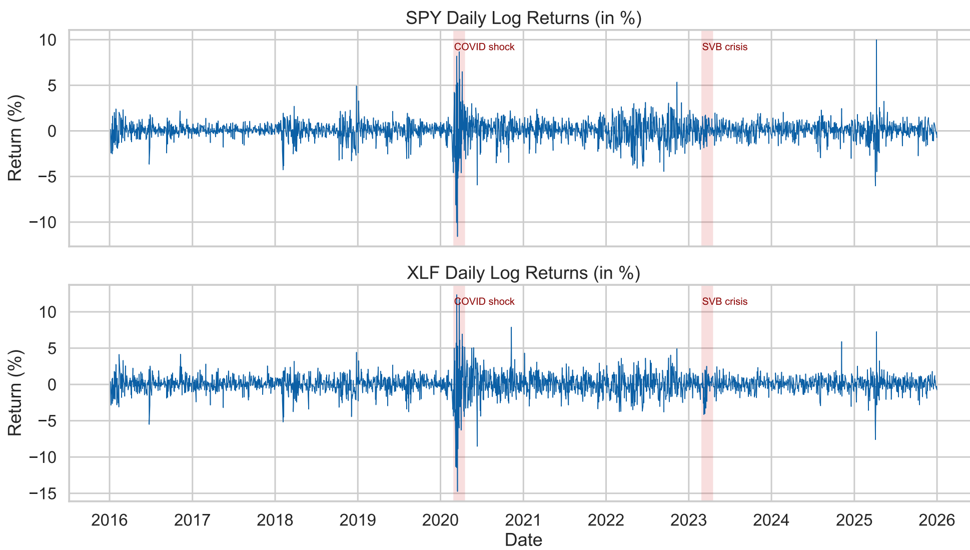

Figure 1. SPY and XLF Daily Returns with Highlighted Stress Periods (COVID 2020 and SVB 2023).

4.2 Key Parameter Evidence

The asymmetric terms are statistically significant in all specifications. For GJR-GARCH, the leverage parameter is positive and highly significant for both assets: $\gamma_{\mathrm{SPY}} = 0.2002$ ($p < 0.001$) and $\gamma_{\mathrm{XLF}} = 0.2024$ ($p < 0.001$). For EGARCH, the signed asymmetry term is negative and highly significant: $\gamma_{\mathrm{SPY}} = -0.1448$ and $\gamma_{\mathrm{XLF}} = -0.1365$, both with very small p-values.

These estimates provide robust evidence of bad-news amplification. XLF shows at least comparable and often stronger asymmetry significance than SPY, consistent with the economic narrative that financial stocks react more sharply under downside stress.

4.3 Information Criteria Comparison

Table 2 reports in-sample fit diagnostics.

EGARCH provides slightly lower AIC/BIC for both assets, indicating marginally better in-sample likelihood fit.

Table 2. Information Criteria by Asset and Model (In-Sample)

Ticker	Model	AIC	BIC	LogLik
SPY	GJR-GARCH	5146.0073	5174.0368	-2568.0037
SPY	EGARCH	5128.8780	5156.9075	-2559.4390
XLF	GJR-GARCH	6343.1126	6371.1420	-3166.5563
XLF	EGARCH	6341.6990	6369.7285	-3165.8495

5 Risk Management Application: VaR & ES

5.1 Out-of-Sample Rolling Forecast Setup

The last 20% of observations (approximately 2024–2025) are reserved for out-of-sample evaluation. A one-step-ahead fixed-window rolling procedure generates time-varying volatility forecasts $\hat{\sigma}_{t+1}$ and corresponding dynamic VaR/ES at 95% and 99%.

5.2 Dynamic Risk Boundary Behavior

Figure 2 illustrates the expected risk-system behavior: during high-volatility episodes, the estimated lower-tail VaR bands move downward (become more conservative), widening the protective buffer against large losses.

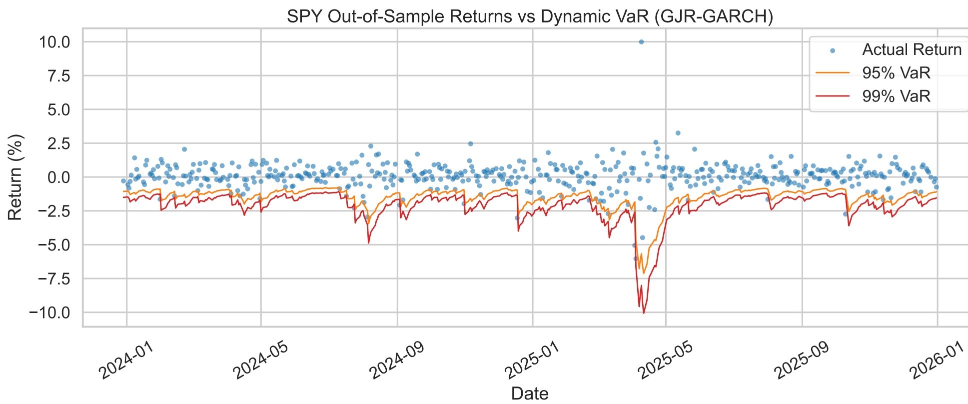

Figure 2. SPY Out-of-Sample Returns with Dynamic 95% and 99% VaR (GJR-GARCH).

To explicitly incorporate tail severity (not only tail quantiles), Figure 3 overlays ES alongside VaR. As expected, ES is always more conservative (more negative) than VaR at the same confidence level, and the ES bands widen further during volatility bursts.

5.3 Backtesting via Exceptions

Table 3 compares observed versus expected exception frequencies.

At 99%, GJR-GARCH has lower absolute calibration error than EGARCH for both SPY and XLF, which is operationally important for stress-capital

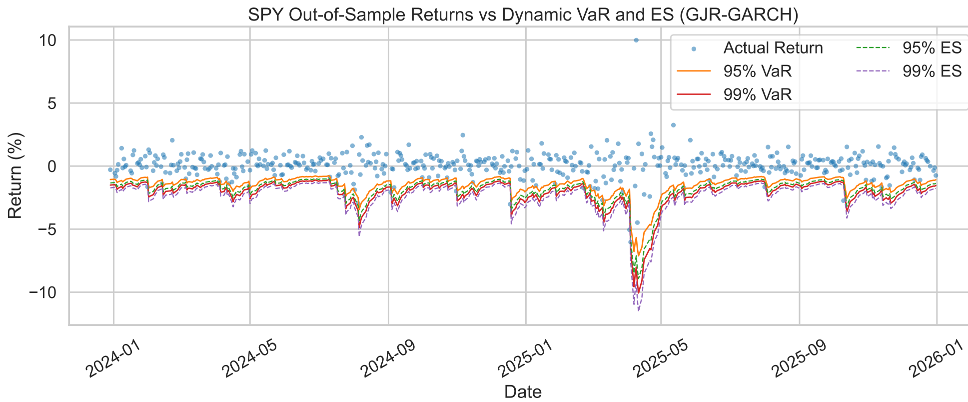

Figure 3. SPY Out-of-Sample Returns with Dynamic VaR and ES (GJR-GARCH).

Table 3. VaR Exception Backtesting Summary (Out-of-Sample)

Ticker	Model	VaR Level	Expected Rate	Observed Rate	Abs. Error
SPY	GJR-GARCH	95%	0.0500	0.0497	0.0003
SPY	GJR-GARCH	99%	0.0100	0.0199	0.0099
SPY	EGARCH	95%	0.0500	0.0497	0.0003
SPY	EGARCH	99%	0.0100	0.0239	0.0139
XLF	GJR-GARCH	95%	0.0500	0.0378	0.0122
XLF	GJR-GARCH	99%	0.0100	0.0179	0.0079
XLF	EGARCH	95%	0.0500	0.0457	0.0043
XLF	EGARCH	99%	0.0100	0.0199	0.0099

decisions and downside limit setting. At 95%, SPY is effectively tied across models while EGARCH performs better for XLF.

5.4 ES Backtesting and Tail-Loss Accuracy

ES is assessed on VaR-breach days by comparing realized tail mean losses against the model-implied ES mean. Table 4 reports the gap

$$\text{ES Gap} = \overline{r_t \mid r_t < \text{VaR}} - \overline{\text{ES}_t \mid r_t < \text{VaR}}. \tag{10}$$

Values below zero indicate realized losses are more severe than predicted ES (underestimation of tail severity).

The ES results indicate that both models underestimate extreme tail severity on breach days (all ES gaps are negative). In relative terms, EGARCH is slightly closer at SPY 99% and XLF 95%, while GJR-GARCH is closer at SPY 95% and XLF 99%. Combined with VaR exception calibration, this suggests a practical hybrid interpretation: GJR-GARCH is preferable for 99% quantile control, whereas ES severity diagnostics should still be monitored model-by-model.

Table 4. ES Backtesting Summary on VaR-Breach Days (Out-of-Sample)

Ticker	Model	ES Level	Tail Obs	Realized Tail Mean	Predicted ES Mean
SPY	GJR-GARCH	95%	25	-2.0963	-1.7581
SPY	GJR-GARCH	99%	10	-2.3606	-1.7891
SPY	EGARCH	95%	25	-2.0963	-1.7623
SPY	EGARCH	99%	12	-2.6115	-2.2433
XLF	GJR-GARCH	95%	19	-2.6258	-2.1766
XLF	GJR-GARCH	99%	9	-3.4286	-3.0286
XLF	EGARCH	95%	23	-2.3819	-2.0181
XLF	EGARCH	99%	10	-3.2706	-2.8414

6 Conclusion

This study compares GJR-GARCH and EGARCH for dynamic market-risk measurement in SPY and XLF. Three conclusions emerge. First, both assets display strong non-normality and asymmetric volatility dynamics, validating the use of asymmetric conditional-variance models. Second, EGARCH slightly dominates in-sample information criteria. Third, and most relevant for risk governance, GJR-GARCH provides better 99% VaR calibration in out-of-sample backtesting for both assets.

For practical deployment, institutions should prioritize model performance at extreme confidence levels and complement VaR with ES for tail severity control. In Basel-oriented settings and banking-sector stress regimes, 99% ES offers a more conservative and policy-aligned safety layer than standalone 95% VaR thresholds.

7 Appendix: Replication Instructions

Environment

Python 3.11 in Conda environment `myenv`. Core packages: `pandas, numpy, yfinance, arch, scipy, matplotlib, seaborn`.

Project Files

Current implementation is script-centric:

- `codes.py`: full pipeline (download, estimation, forecasting, VaR/ES, plotting, exports)
- `outputs/data`: return and VaR/ES time-series CSV files
- `outputs/tables`: descriptive, estimation, IC, and backtesting tables
- `outputs/tables/es_backtest_summary.csv`: ES tail-loss evaluation table
- `outputs/figures`: Figure 1, VaR overlays (Figure 2), and VaR+ES overlays (Figure 3)

Reproducibility and Execution Guide

1. Go to the project folder: `Assignment Answers by Ruiwen HE/FINAL PAPER`.

2. Run the full pipeline:

```
D:/Programming/Python/Conda/envs/myenv/python.exe codes.py
```

3. Review generated artifacts in:

   `outputs/data` (time-series data), `outputs/tables` (empirical tables), and `outputs/figures` (publication figures).